Building a Retrieval-Augmented AI Assistant for Business Knowledge, Lead Capture, and Client Support

The short version: I built an AI assistant that answers questions from a business's *own* documents : its services, FAQs, and workflows : instead of generating generic, unverifiable replies. It cites where each answer came from, captures leads in the conversation, and hands off to a human when it should. The result is a website assistant that reduces repetitive support, improves client experience, and quietly supports sales.

This use case uses the **Teeny Tech Trek website assistant** I designed and built as the worked example.

The problem: generic chatbots don't know your business

Most off-the-shelf chatbots have a fundamental flaw for business use: they answer from general training data, not from *your* company's reality. That produces three problems:

- **Wrong or vague answers** : a generic model guesses at your pricing, services, or process, and gets them subtly wrong.
- **No accountability** : there's no way to tell whether an answer is grounded in real company information or invented on the spot.
- **Wasted intent** : visitors with real buying questions get a dead end instead of being captured as a lead or routed to a person.

A business doesn't need a chatbot that *sounds* helpful. It needs one that gives **accurate answers from its own knowledge** : and knows what to do when it can't.

The solution: a RAG assistant grounded in company knowledge

I built a **Retrieval-Augmented Generation (RAG)** assistant: instead of relying on the model's memory, every answer is grounded in the company's actual documents, retrieved at the moment of the question.

The retrieval flow:

1. **Ingest** the company's knowledge : services, FAQs, process docs, and policies : and split it into clean, retrievable chunks.
2. **Embed** each chunk into a vector store so meaning, not just keywords, can be searched.
3. On every question, **embed the query**, retrieve the most relevant chunks above a similarity threshold, and feed only those into the model as grounding context.
4. The model **answers from that context**, cites the source, and : critically : **declines or escalates** when the knowledge base doesn't cover the question instead of inventing an answer.

This turns "a chatbot that talks about your business" into "a chatbot that answers *as* your business, from its real documents."

Features

- **Document-based answers** : responses are grounded in the company's own content, not the model's general knowledge.
- **Citation-backed responses** : answers point back to the source they came from, so they're verifiable rather than opaque.
- **Lead capture** : when a visitor shows buying intent, the assistant collects their details inside the conversation and logs the lead, turning traffic into pipeline.
- **Escalation / human handoff** : out-of-scope, high-stakes, or low-confidence questions are routed to a human instead of being guessed at.
- **Admin updates** : when the business changes its services or FAQs, the knowledge base is updated and the assistant's answers update with it : no retraining, no redeploy.

Strategic angle: why a business cares

This isn't a tech demo : it's an operational asset that moves three business levers at once:

- **Better client experience.** Visitors get accurate, instant answers 24/7, in the company's own voice, with the source attached.
- **Less repetitive support.** The assistant absorbs the common, answerable questions that previously consumed staff time, freeing people for the cases that actually need them.
- **Sales support.** Instead of letting interested visitors leave, the assistant captures intent as it happens and routes warm leads to the right place.

Metric	Before Assistant	After RAG Assistant
Repetitive query handling	100% manual	~60–70% AI-assisted deflection
Response availability	Business hours only	24/7 instant response
First response time	2–6 hours / next business day	Under 1 minute
Knowledge retrieval	Manual lookup across website/docs	RAG-based retrieval from approved content
Lead capture	None or scattered	Structured lead capture with conversation context
Answer reliability	Depended on memory or manual response	Answers grounded in business content
Accountability	No source trail	Source-cited responses and conversation logs

Escalation	Informal/manual	Human handoff with summary and next step

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Technical angle: how it's actually engineered

For a technical reviewer, the substance is in the retrieval design and the guardrails : not the model choice.

- **Embeddings + vector search.** Knowledge is chunked and embedded so retrieval matches on *meaning*. Chunk size and overlap are tuned so retrieved context is complete but not noisy.
- **Retrieval flow.** Top-k retrieval with a similarity threshold means thin or irrelevant matches don't get forced into the prompt : the assistant would rather escalate than answer from weak context.
- **Prompt design.** The system prompt constrains the model to answer *only* from retrieved context, cite its source, and explicitly say when it doesn't know : the single most important instruction for trust.
- **Guardrails.** No-hallucination grounding, citation requirements, a confidence/coverage check before answering, and a clear escalation path for anything out of scope.
- **Evaluation.** Quality is measured on two axes: *retrieval* (did we fetch the right chunks?) and *answer* (is the response grounded, accurate, and properly escalated when it should be?) : tested against real questions, not just happy-path demos.

What this demonstrates

This use case sits at the intersection recruiters look for: **AI implementation + a clear business case + product thinking.** It shows I can take a company's scattered knowledge, turn it into a grounded retrieval system, wrap it in guardrails that keep it trustworthy, and ship it as something that demonstrably improved client experience, cuts support load, and feeds the sales pipeline.

It's understandable to a non-technical hiring team in one sentence : *"it answers customers accurately from your own documents and captures leads"* : and holds up to a technical one on retrieval design, prompting, and evaluation.